Problem Statement

1. Using Cerebrum Weil crate to write agents including human-in-the-loop

2. Use external agentic framework like Langchain or Google ADK and add Weilliptic audit logging into the mix

3. Develop e-commerce application, deploy on Weilchain and leverage stable coin infrastructure of the Weilchain

Solution/ Hackathon Idea:-

Autonomous Dropshipping Agent on WeilChain

Overview

This project implements an **AI-powered autonomous dropshipping system** that integrates **AI agents, a Rust-based StepAgent deployed on WeilChain, and a web-based interface** to automate supplier selection, order approval, and payment workflows.

The system demonstrates how **intelligent agents and blockchain-based workflow execution** can automate e-commerce operations while maintaining **human oversight and auditability**.

The project consists of three major components:

1. **Frontend (React + Vite)** – Customer and Admin dashboards
2. **Backend (FastAPI + Python AI agent)** – Order orchestration and supplier selection
3. **Rust StepAgent (WeilChain)** – Secure workflow execution and approval logic

System Architecture

Customer places order

↓

AI agent selects supplier

↓

Rust StepAgent begins workflow

↓

Cart review

↓

Risk analysis

↓

Human approval required

↓

Customer pays using WUSD

↓

Supplier order placed

↓

Fulfillment update

Core Components

1 Frontend (React + Vite)

The frontend provides two interfaces:

Customer Store

Customers can:

- Browse products
- Add items to cart
- Create an order
- Pay for approved orders

Admin

The admin dashboard allows a human reviewer to:

• view pending orders
• review supplier selection
• approve or reject the order

This is where the **human-in-the-loop approval occurs**.

Backend (FastAPI)

The backend acts as the **orchestrator of the entire workflow**.

Responsibilities:

• Receive cart data from frontend
• call the AI supplier agent
• start the Rust StepAgent workflow
• track order states
• handle human approvals
• generate payment intents
• confirm payments

AI Supplier Selection Agent (Python)

The supplier agent evaluates available suppliers and selects the best one based on price and shipping cost.

The agent calculates and chooses the supplier with lowest total cost

This introduces an AI-based decision step in the workflow

Rust Cerebrum StepAgent (WeilChain)

The **core workflow logic runs inside a Rust smart contract agent** deployed on WeilChain using the **Cerebrum agent framework**.

This agent executes a **multi-step workflow state machine**.

Workflow steps:

CartReview
RiskCheck
HumanApproval
PayWithWUSD
PlaceSupplierOrder
FulfillmentUpdate

Human-in-the-Loop Approval

The human approval mechanism is implemented using:

```
RunStatus::Pending
```

When the workflow reaches the **HumanApproval step**, the agent pauses execution.

```
HumanApprovalState → RunStatus::Pending
```

This signals that the workflow must wait for external input.

The admin then reviews the order and clicks **Approve**.

Payment Using WUSD Stablecoin

After approval, the system creates a **payment intent**.

The customer then pays using **WUSD**, the stablecoin on WeilChain. Once payment is confirmed the rust agent continues the workflow

Technologies Used

Frontend

• React
• TypeScript
• Vite

Backend

• FastAPI
• Python
• REST APIs

AI Logic

• custom supplier decision agent

Blockchain Layer

• WeilChain
• Cerebrum agent framework
• Rust smart contract

Payments

• WUSD stablecoin